

Extending the AI Assistant

From Conversational Analytics to an Agentic AI Platform
across ERP Financials, Portfolio Management & HR

Scope	ERP (Financials) · PMS (Portfolio / Stocks) · HR (Basic) · Mobile
Focus	NSE/BSE market data · PMS intelligence (Buy/Sell/Hold, what-if, predictive) · Agentic AI · Agent c
Status	Proposal for review

1. Executive Summary

We have already delivered a working AI Assistant inside the ERP: it understands the full database schema, converts business questions into safe SQL through a guarded gateway, executes them live, and answers with real numbers — every query logged and access-controlled. Business pages (accounting status dashboards, journal previews, customer inquiries) already reuse the same AI data path.

This proposal extends that foundation in three directions: **(1) external data connectivity** — the assistant reaches out to NSE / BSE and other market-data services so portfolio questions combine internal PMS holdings with live market prices; **(2) an agentic architecture** — instead of only answering questions, the AI runs *agents*: goal-driven workers with tools, schedules, and approval gates that carry out multi-step work in ERP and PMS; **(3) a user agent builder** — power users compose their own agents from approved tools without writing code.

The result is a single AI layer over ERP, PMS and HR that answers, monitors, reconciles, alerts, drafts and — with human approval — acts.

2. What the AI Assistant Does Today

- **Natural language to SQL** over the complete ERP schema (GL, AP, AR, FA, Cash, Petty Cash) via a guarded, SELECT-only SQL gateway — single statement, keyword bans, row caps, full audit log.
- **Live schema catalog** with table/column comments and an access-control list, refreshed on demand, so answers always reflect the current database.
- **Conversational analysis** with charts, drill-downs, image and PDF attachments the assistant can read.
- **Saved AI reports and scheduled jobs** — reusable, parameterised questions.
- **AI-powered application pages** — Check Accounting dashboard, journal preview, customer accounts — all running on the same transparent SQL path with the query visible to the user.

Everything below builds on this: the same gateway, audit trail and security model become the foundation of the agent platform.

3. The Vision — One AI Layer over Three Systems

ERP (financials back end). The system of record: ledgers, journals, payables, receivables, assets, cash. The AI's role here is assurance and automation — closing periods, reconciling, spotting anomalies, preparing postings for approval.

PMS (portfolio management — the priority). Holdings, trades, positions and stock data live here. The AI's role is intelligence: valuing portfolios with live NSE/BSE prices, monitoring risk and limits, tracking corporate actions, generating research summaries and alerts, and reconciling PMS positions against ERP financials.

HR (basic). Employee master, leave, payroll basics. The AI's role is service: policy questions, leave balances, payroll sanity checks — low risk, high convenience.

One assistant, one security model, one audit trail — three systems. A user asks *“what is my exposure to banking stocks and how did it move against the ledger this month?”* and the AI joins PMS holdings, NSE prices and GL balances in one answer.

4. External Data Connectivity (PMS)

A **connector layer** sits beside the internal SQL gateway. Each connector is a typed, rate-limited, cached adapter the AI can call as a tool — the same way it calls SQL today. All calls are logged with who / what / when, and responses are cached so repeated questions do not hammer external services.

Planned connectors

Agent	What it does	Data / tools	Actions it can take	Autonomy
NSE / BSE Quotes	Live and delayed equity quotes, indices, market depth for symbols held in the PMS.	Exchange / licensed data vendor APIs	Read-only fetch; cache with TTL per symbol	Automatic
EOD Prices (Bhavcopy)	Official end-of-day prices and volumes for all listed symbols; drives daily MTM valuation.	NSE/BSE end-of-day files	Scheduled nightly import into a price store	Automatic
Corporate Actions	Dividends, splits, bonuses, rights, mergers affecting held symbols.	Exchange announcements feeds	Detect impact on holdings; draft adjustment entries	Notify + approve
Indices & Benchmarks	NIFTY/SENSEX and sector indices for performance comparison.	Exchange index feeds	Read-only fetch	Automatic
FX Rates	Currency rates for multi-currency valuation and ERP revaluation cross-check.	Central bank / rate service	Read-only fetch	Automatic
News & Filings	Company announcements and news for held symbols, summarised by the AI.	Exchange filings, news APIs	Summarise; attach to holdings; alert on keywords	Automatic

What this unlocks immediately

- **Live portfolio valuation** — mark-to-market of every PMS position at current or EOD prices, by portfolio, sector, or client.
- **Performance vs benchmark** — portfolio return against NIFTY/SENSEX over any window.
- **Exposure analysis** — concentration by stock, sector, market cap; what-if moves.
- **Price alerts** — “tell me when any holding falls 5% in a day” becomes a standing agent.
- **PMS ↔ ERP tie-out** — investment balances in the GL reconciled against valued PMS positions.

5. The PMS Intelligence Engine — Buy / Sell / Hold Decision Support

The heart of the PMS extension: for every holding (and every watched symbol) the system continuously maintains an **investment view** — a scored, explained recommendation of **Buy / Add, Hold, Reduce, or Sell** — that the portfolio manager can interrogate, stress and act on. The AI prepares the decision; the manager takes it. Every recommendation carries its full reasoning chain and the data it was computed from.

5.1 The signal stack — what feeds a recommendation

Agent	What it does	Data / tools	Actions it can take	Autonomy
Fundamental signals	Earnings growth, margins, debt, ROE, valuation multiples vs sector and history — from published financials of listed companies.	Company financials feeds, exchange filings	Quarterly refresh; re-score on new results	Automatic
Technical signals	Trend (moving averages), momentum (RSI, MACD), support/resistance, volume behaviour, relative strength vs index.	Price/volume history store	Daily recompute after EOD import	Automatic
Risk signals	Volatility, beta, drawdown, correlation to the rest of the book, liquidity (days-to-exit at average volume).	Price history, portfolio positions	Daily recompute	Automatic
News & sentiment	AI-read announcements, results commentary, analyst actions and news flow per symbol, scored for direction and materiality.	News/filings connectors, Claude analysis	Continuous; material events raise alerts	Automatic
Portfolio context	Position size vs limits, cost basis and tax status, concentration effect, cash available, mandate constraints.	PMS positions, compliance rules	On every evaluation	Automatic

5.2 The decision layer

- **Composite scoring** — each signal group produces a normalised score; a transparent, configurable weighting (the client's investment policy, not a black box) combines them into the Buy/Hold/Reduce/Sell view with a confidence level.
- **Written rationale** — the AI writes the case in plain language: what supports the view, what argues against it, what would change it (“downgrade to Reduce if results miss on margins”). Managers debate with it in chat.
- **Action drafts** — an accepted recommendation becomes a draft trade sized within limits, routed through the approval inbox like every other write.
- **Recommendation journal** — every view, its inputs and its outcome are stored, so the system's advice is measurable over time (hit rate, average gain per call) and auditable.

5.3 What-if & scenario analysis

- **Trade what-ifs** — “what if I sell half of X and buy Y?”: instant recomputation of exposure, sector weights, expected yield, limit compliance and realised P&L/tax impact before any order exists.
- **Market scenarios** — shock the book: index $\pm 5/10\%$, sector shocks, rate moves, single-stock crash; see portfolio value, margin of safety and which limits would break.

- **Historical replays** — replay the current portfolio through past episodes (e.g. a crash month, an election week) using stored price history.
- **Goal seeking** — “what changes get me to 20% banking exposure with minimum turnover?” — the optimizer proposes the smallest trade set.

5.4 Predictive analytics

- **Price & return forecasting** — statistical/ML models (trend, mean-reversion, seasonality) produce forward return distributions with confidence bands — always shown as ranges, never single-point promises.
- **Volatility & drawdown forecasting** — expected volatility per holding and for the book; probability of breaching a drawdown threshold over the horizon.
- **Earnings-event awareness** — flags upcoming results/dividend dates for holdings and the historical price behaviour around them.
- **Cash-flow prediction** — expected dividends and maturities feeding the ERP cash forecast.
- **Backtesting** — any rule or the composite scorer itself can be run over history to show how it would have performed before it is trusted live.
- **Model governance** — every model is versioned, its accuracy tracked against outcomes, and retrained on schedule; predictions carry the model version that produced them.

Compliance position: the engine is decision *support*. Recommendations are advisory, human-approved, logged and explainable — aligned with the discretionary-management model and ready for regulatory review of how advice is produced.

6. Agentic AI Architecture

An **agent** is a goal + instructions + a set of allowed tools + a trigger (chat, schedule, or event) + an approval policy. The platform provides the runtime; agents are configuration, not code — which is what makes the user-built agents in section 9 possible.

Platform components

Agent	What it does	Data / tools	Actions it can take	Autonomy
Orchestrator	Runs the agent loop: plan → call tools → observe → continue or finish. Handles retries, timeouts, budgets (max steps / tokens).	Claude models via API	—	—
Tool Registry	Catalog of everything an agent may use: guarded SQL (read), ERP web services (write), market-data connectors, e-mail/notification, report/Excel generator.	Existing ORDS endpoints + connectors	Each tool declares read or write and its approval class	—
Scheduler	Cron-style and event triggers (period opened, file arrived, price moved). Runs agents unattended within their permissions.	Database scheduler + event hooks	Start agent runs	—
Approval Gate	Any write action (post journal, create payment, adjust position) is drafted by the agent and queued for a named human to approve in-app before execution.	In-app approval inbox	Hold / approve / reject with comments	Human-in-the-loop
Audit Trail	Every agent step — prompt, tool call, SQL, result, approval — is logged, extending today's AI query log.	AI action log tables	Immutable record; reviewable per run	—
Guardrails	Schema ACL, SELECT-only SQL, per-agent tool allow-lists, spend/step budgets, PII masking, no external calls except registered connectors.	Policy engine	Block + report violations	—

Autonomy levels

- **Answer** — read-only; responds with data and analysis (default for all agents).
- **Notify** — watches on a schedule and pushes alerts/digests; still read-only.
- **Draft + Approve** — prepares the transaction (journal, payment batch, adjustment) and routes it to a human; nothing posts without sign-off.
- **Auto within policy** — executes without a person only for whitelisted, low-risk, reversible actions (e.g. refreshing a price cache), and only where the client explicitly enables it.

7. Agent Catalog — ERP / Financials

Agent	What it does	Data / tools	Actions it can take	Autonomy
Period Close Agent	Walks the month-end checklist: all subledgers accounted, journals posted, reconciliations clean, retained earnings rolled. Produces a close-readiness scorecard.	GL, AP, AR, FA, Cash tables; accounting-status views	Run checks; open exceptions list; draft closing journals	Notify / Draft
Accounting Status Agent	Automates today's Check Accounting dashboard: every document vs its GL journal, per period and module, with drill-down to the offender.	Accounting status views; GL reference matching	Alert on unaccounted items; create accounting for approved docs	Notify / Draft
Bank Reconciliation Agent	Matches bank statement lines to external transactions and GL; proposes matches for the rest by amount/date/reference similarity.	Bank statements, external transactions, GL	Auto-match exact; propose fuzzy matches for confirmation	Draft + Approve
AP Invoice Intake Agent	Reads supplier invoices (PDF/scan) with vision, extracts header and lines, matches PO/receipt, checks duplicates and totals, drafts the invoice entry.	Attachments, supplier master, PO data	Create draft invoices; flag mismatches	Draft + Approve
Payment Run Agent	Builds payment proposals: due invoices, discounts expiring, cash available, duplicate/blocked supplier checks.	AP invoices, payment terms, bank balances	Draft payment batch for approval; predict cash need	Draft + Approve
Collections Agent	Ranks overdue receivables, drafts reminder e-mails per customer with statement attached, tracks promises to pay.	AR invoices, receipts, customer master	Send approved dunning e-mails; log follow-ups	Draft + Approve
Cash Forecast Agent	Projects 13-week cash: AP due, AR expected (weighted by payment history), payroll, recurring items.	AP/AR open items, bank balances, history	Publish weekly forecast; alert on shortfalls	Notify
Journal Sentinel	Watches new journals for anomalies: round-sum entries, weekend postings, unusual accounts pairs, out-of-balance batches, back-dated entries.	GL journals, posting history baselines	Flag to controller with explanation	Notify
Fixed Assets Agent	Runs monthly depreciation previews, posts approved runs, tracks retirements and their accounting, reconciles FA register to GL.	FA registers, depreciation detail, GL	Draft depreciation postings; retirement checklists	Draft + Approve
Revaluation & RE Agent	Month-end FX revaluation of open balances and retained-earnings rollforward, using the rules already built into the ERP.	TB, FX rates, RE tables	Draft revaluation journals; save RE calculations	Draft + Approve

Agent	What it does	Data / tools	Actions it can take	Autonomy
Report Writer	Turns a request — “board pack for Q1” — into formatted Excel/PDF: TB, P&L;,, aging, commentary generated from the numbers.	All financial data; report templates	Generate and distribute on schedule	Notify
Master Data Steward	Finds duplicate suppliers/customers, missing account assignments (e.g. BU receivables accounts), inactive-but-used records.	Supplier/customer/account masters	Propose merges and fixes for approval	Draft + Approve

8. Agent Catalog — PMS (Portfolio / Stocks)

Agent	What it does	Data / tools	Actions it can take	Autonomy
Market Data Agent	Keeps the price store fresh: intraday quotes for held symbols, nightly EOD bhavcopy import, index levels, FX.	NSE/BSE connectors, price store	Fetch, validate, cache; alert on stale/failed feeds	Automatic
Valuation (MTM) Agent	Values every portfolio daily at EOD (or on demand intraday): position × price, cost basis, unrealised P&L; by portfolio/client/sector.	PMS positions, price store	Publish valuation snapshots; feed ERP investment balances	Notify
Risk & Exposure Agent	Concentration (single stock, sector, group), volatility of the book, drawdown tracking, scenario questions ("if NIFTY -3%?").	Positions, price history, indices	Daily risk digest; threshold alerts	Notify
Compliance / Limits Agent	Enforces mandate rules: max % per stock/sector, restricted lists, min cash. Checks pre-trade (assistant) and post-trade (scheduled).	Positions, limit rules, trades	Block-list warnings; breach reports with deadlines	Notify
Corporate Actions Agent	Detects dividends/splits/bonuses on held symbols, computes entitlement, adjusts positions and drafts the ERP income entries.	Corporate action feeds, positions	Draft position adjustments + GL entries	Draft + Approve
Trade Capture Assistant	Converts broker contract notes (PDF) into trade records: parse, validate against orders, compute charges, post to PMS and draft settlement entries.	Contract notes, broker masters	Create draft trades; settlement journal drafts	Draft + Approve
Performance Agent	Time-weighted returns per portfolio vs benchmark; attribution by stock and sector; monthly client-ready factsheets.	Valuation history, benchmarks	Generate factsheets; explain performance drivers	Notify
Rebalancing Advisor	Compares actual vs model allocation; proposes trades to rebalance within compliance limits and tax/charge awareness.	Positions, model portfolios, prices	Draft trade list for the manager to review	Draft + Approve
Research Summarizer	Watches filings/news for held symbols; produces a morning brief: what happened, which holdings are affected, sentiment.	News/filings connectors	Morning digest; ad-hoc "what's new on X?"	Notify
Price Alert Agent	User-defined standing alerts: price/percentage moves, 52-week levels, volume spikes on held or watched symbols.	Live quotes, user watchlists	Push notification / e-mail on trigger	Automatic

Agent	What it does	Data / tools	Actions it can take	Autonomy
PMS-ERP Reconciler	Ties PMS holdings and cash to the ERP investment and bank GL accounts; explains every difference (pending settles, fees, FX).	PMS positions, GL balances	Daily tie-out report; draft correcting entries	Draft + Approve

9. Agent Catalog — HR (Basic)

Agent	What it does	Data / tools	Actions it can take	Autonomy
HR Helpdesk	Answers policy and personal questions — leave balance, holiday calendar, benefits — scoped to the asking employee's own data.	HR tables, policy documents (RAG)	Answer; open a ticket when out of scope	Answer
Leave & Attendance Agent	Monitors pending approvals, negative balances, absence patterns; reminds approvers.	Leave/attendance records	Reminders; monthly attendance summary	Notify
Payroll Verifier	Pre-payroll sanity: headcount vs last month, unusual variances per employee, missing bank details, GL payroll tie-out.	Payroll run data, GL	Variance report before payroll approval	Notify
Onboarding Agent	Checklist runner for joiners/leavers: accounts, assets assigned/returned, document collection status.	HR master, asset register	Chase owners of open items	Notify

10. The Agent Builder — Users Create Their Own Agents

A no-code screen inside the application where an authorised user assembles an agent in five steps:

- **1. Name & goal** — plain-language instructions: “Every morning check yesterday's receipts above 100k and summarise by customer.”
- **2. Tools** — tick the tools it may use from the registry; the user can never grant a tool beyond their own role.
- **3. Trigger** — chat-only, schedule (cron picker), or event (period opened / file uploaded / price moved).
- **4. Approval policy** — read-only, notify, or draft-for-approval with a named approver.
- **5. Test & publish** — dry-run with visible tool calls (the same transparency the SQL pages have today), then publish privately or to the team.

Every published agent is versioned, owned, auditable and can be paused instantly. The agents in sections 6–8 ship as pre-built templates in the same framework — the client's teams then clone and adapt them.

11. AI Forms Designer — From Conversation to Application

The chatbot is where users discover which analysis matters; the Forms Designer is how that analysis becomes a permanent, refreshing screen. When an answer in chat is worth keeping, the user says “*make this a form*” — and describes, in natural language, how it should look and refresh. The AI generates the form; from then on the user monitors the data on a structured page instead of re-asking the chatbot.

How it works

- **1. Pin an analysis** — from any chat answer, saved report or agent result: “pin this as a form”.
- **2. Describe the form** — natural language: “KPI tiles on top for total overdue and count; below, the grid by customer; red when overdue > 30 days; refresh every hour; add a period selector.”
- **3. AI generates the form spec** — a declarative layout of blocks, each bound to a data source: a guarded SQL query, an **agent's latest result**, or a market-data connector. The spec is stored, versioned and editable — no code.
- **4. Publish** — the form appears in the application menu (and on mobile), shareable with chosen roles; every block re-checks the viewer's data permissions, not the author's.
- **5. Edit by conversation** — the form has its own chat: “add a column for days overdue”, “split by business unit”, “alert me when the total passes 1M” — each request updates the spec, with history.

Building blocks

- **KPI tiles** — single numbers with trend arrows and thresholds (green/amber/red).
- **Data grids** — sortable, filterable, exportable to Excel; drill-through to the source record pages.
- **Charts** — trends, compositions, comparisons, generated from the bound query.
- **Agent panels** — the structured output of any agent run embedded live: the Accounting Status agent's exceptions, the Risk agent's daily digest, the Collections agent's top-ten list — several agents on one form.
- **Parameter bar** — period, company, portfolio, customer selectors that re-drive every block.
- **Narrative block** — an AI-written summary paragraph regenerated on each refresh (“what changed since yesterday”).
- **Input blocks** — forms are two-way: entry fields, dropdowns bound to LOVs, date pickers and file attachments let the user capture data directly on the form — a remark on an exception, a dispute status on an overdue invoice, a decision on a recommendation, a new record.
- **Action buttons** — a button on the form invokes a registered write tool: mark resolved, create the draft journal, start the agent, send the reminder. Writes go through the same validated web services and approval gates as everything else — never raw SQL.

Governance built in

- **Viewer-side security** — blocks execute under the viewer's ACL; sharing a form never shares data access.
- **Cached refresh** — blocks refresh on their schedule into a cache; opening a form reads the cache, so popular forms cost one query per interval, not one per viewer.
- **Lifecycle** — every form has an owner, version history, usage statistics and automatic archiving when unused; an admin console lists all published forms.
- **Write safety** — input blocks and action buttons can only bind to write tools the form owner's role allows; entered data is validated by the target service, attributed to the entering user, and approval-classed actions still stop at the approval inbox.

12. Mobile App — Monitor Everything, Approve Anywhere

A companion mobile app (iOS + Android from one codebase) puts the whole platform in the manager's pocket. It is not a second system: it consumes the same secured APIs, the same agents and the same approval inbox as the web application.

Agent	What it does	Data / tools	Actions it can take	Autonomy
Portfolio Cockpit	Live portfolio value, day P&L; top movers, exposure wheel; drill from total to position to the AI's current Buy/Hold/Sell view with its rationale.	PMS APIs, price store	View; ask the assistant about anything on screen	—
Alerts & Notifications	Push notifications from every agent: price moves, limit breaches, unaccounted documents, failed feeds, close-checklist exceptions.	Agent runtime, push service	Acknowledge, mute, escalate, open drill-down	—
Approvals Inbox	Everything agents drafted — journals, payment runs, trades, corporate-action adjustments — approved or rejected with comment, the full AI rationale attached.	Approval gate APIs	Approve / reject / hold with biometric confirmation	Human-in-the-loop
AI Chat	The same assistant, mobile-first: voice or text questions over ERP and PMS, charts rendered inline, follow-up drills.	Assistant APIs	Read-only answers; hands off to approvals for actions	—
KPI Dashboards	Close progress, cash position, AR aging, portfolio performance vs benchmark — the numbers leadership checks daily — including any published AI form, rendered mobile-first.	Reporting APIs	View; subscribe to a daily digest	—

- **Security** — device enrolment, biometric unlock, short-lived tokens, per-role screens; approvals require step-up authentication; no data at rest beyond a server-revocable cache.
- **Technology** — React Native, sharing logic and API clients with the existing React web app; push via the standard mobile notification services; offline-tolerant read caches.

13. Security & Governance

- **Read/write separation** — reads go through the guarded SELECT-only SQL gateway; writes only through the existing, validated ERP/PMS web services. Agents never get raw DML.
- **Access control** — the existing object ACL governs what the AI can see; per-agent tool allow-lists govern what it can do; user role caps what a builder can grant.
- **Human-in-the-loop** — financial postings, payments and position changes always stop at an approval inbox unless a policy explicitly whitelists them.
- **Complete audit** — extends today's AI query log to every agent step: prompt, tool, SQL/API call, result hash, approver, timestamps. Answer any auditor's "why did this post?" with the full chain.
- **Data boundaries** — external connectors are outbound-only to registered endpoints; internal data sent to the model is minimised and can be masked (client names, salaries) per policy; no training on client data.
- **Budgets & kill-switch** — per-agent step/token/spend budgets, global pause, and per-run cancellation.

14. Technology Approach

The platform continues the current stack: Oracle APEX/ORDS services and the guarded SQL gateway as the data plane, the React application as the surface, and Anthropic's Claude models as the reasoning engine — selected per task:

Workload	Model tier	Why
Interactive assistant, agent orchestration	Claude Sonnet (latest)	Best balance of reasoning quality, speed and cost for tool-use loops
Complex analysis — close scorecards, reconciliation reasoning, research synthesis	Claude Opus (latest)	Deepest reasoning for multi-step financial logic
High-volume, low-latency — alert classification, extraction, routing	Claude Haiku (latest)	Fast and economical for repetitive checks

- **Tool protocol** — tools exposed in an MCP-style typed registry so new connectors (an exchange, a bank, a broker) plug in without touching agent logic.
- **Document intelligence** — the models' native vision reads invoices, contract notes and statements; a document store with retrieval (RAG) grounds policy and procedure answers.
- **Scheduling** — database scheduler drives unattended runs; the app surfaces run history, logs and approvals.

15. Development & Release Roadmap

Delivery runs as a release train: a production release roughly every 6–8 weeks, each one usable on its own, with the PMS intelligence deepening release by release. Workstreams run in parallel — platform/agents, PMS intelligence, mobile, and data engineering.

Release train

Release	Contents	You can now...	Timeline
R1 — Market Data Foundation	NSE/BSE connectors, EOD price store + nightly import, FX & index feeds, live valuation and exposure in the assistant, price alerts. Data QA dashboards for feed health.	Value any portfolio live; ask exposure questions; get price alerts	Weeks 1–6
R2 — Agent Runtime	Orchestrator, tool registry, scheduler, approval inbox, audit trail. First five agents live: Accounting Status, Bank Reconciliation, Valuation, Risk Digest, Collections.	Agents watch ERP & PMS around the clock; humans approve their drafts	Weeks 5–12

Release	Contents	You can now...	Timeline
R3 — PMS Intelligence v1	Signal stack (technical, risk, portfolio context), composite Buy/Hold/Sell scoring with written rationale, what-if trade simulator, recommendation journal, backtesting harness.	Interrogate a scored view on every holding; simulate trades before ordering	Weeks 11–18
R4 — Mobile v1 + Builder	Mobile app (cockpit, alerts, approvals, chat) on iOS/Android; no-code agent builder; AI Forms Designer ; remaining ERP/PMS agent templates; document intake (invoices, contract notes).	Monitor and approve from the phone; teams build their own agents and forms	Weeks 17–24
R5 — PMS Intelligence v2	Fundamental & sentiment signals, predictive models (return/volatility/drawdown forecasts with confidence bands), scenario & stress engine, rebalancing optimizer, model-governance console.	Predictive, scenario-tested decision support with measured accuracy	Weeks 23–32
R6 — HR + Hardening / GA	HR agents, PII masking, budgets & admin console, performance hardening, penetration test, DR runbook, full documentation and training.	Organisation-wide general availability	Weeks 31–36

How we build and ship

- **Cadence** — two-week sprints inside each release; sprint demos to the client; a hardening sprint before every production cut.
- **Environments** — DEV → TEST (client UAT) → PROD, with automated deployment of database patches, services and frontend; every DB change is a numbered, re-runnable script (the discipline already in use).
- **Quality gates** — typed API contracts, regression suite on the SQL gateway and agent tools, golden-file tests for financial calculations, backtest thresholds a model must beat before it ships, and UAT sign-off per release.
- **Pilot-first** — every capability goes live on one pilot portfolio / one company first, runs in shadow mode (agents observe and report but do not draft) for two weeks, then widens.
- **Team shape** — 1 architect/lead, 2 full-stack (web + services), 1 data engineer (feeds/price store), 1 quant/ML for the intelligence engine, 1 mobile developer (from R3), 1 QA. Client side: product owner, a portfolio manager as domain expert, and a finance approver.
- **Release management** — versioned releases with change logs, feature flags for gradual enablement, one-click rollback of frontend and services, database rollback scripts prepared with every patch.

16. Business Benefits & Success Measures

- **Faster close** — accounting-status and reconciliation agents surface exceptions daily instead of at month-end; target: days off the close cycle.
- **Portfolio decisions on live data** — valuation, exposure and news in one place, minutes not days; target: same-day MTM every day.
- **Fewer errors and leakage** — duplicate invoices, limit breaches, unaccounted documents caught by standing agents; target: measurable drop in exceptions aged > 7 days.
- **Team leverage** — routine preparation (payment runs, dunning, factsheets) drafted by agents, humans approve; target: hours returned per week per team.
- **Full transparency** — every AI answer and action traceable to the exact query and approval; audit-ready by construction.

17. Proposed Next Steps

- Workshop with the client to rank the agent catalog and confirm the Phase-1 market-data scope (symbols, exchanges, licensing of data feeds).
- Confirm the market-data vendor / exchange access route and commercial terms for NSE/BSE feeds.

- Approve the agent security model (approval matrix, autonomy levels, masking policy).
- Kick off Phase 1 with a two-week spike: NSE/BSE connector + live valuation of one real portfolio as the demonstrable proof.

— End of proposal —